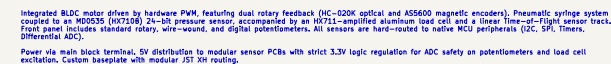


STM32F407



Wireless co-processor. Interfaces directly with the STM32 via a serial UART connection (PB10/PB11) to handle external telemetry, wireless data logging, or network communication.



Power via main block terminal, 5V distribution to modular sensor PCBs with strict 3.3V logic regulation for ADC safety on potentiometers and load cell excitation, Custom baseplate with modular JST XH routing.

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Main power distribution network. Converts raw input voltage into stable 5V and 3.3V logic rails via dual regulators for the MCU and sensors, while providing direct high-current passthrough to the ESC and motor terminals.

Galvanically isolated output stage. Employs optocouplers driven by the STN32 (PD12, PD13, PD14) to safely trigger external industrial hardware without exposing the logic board to voltage spikes or ground loops.

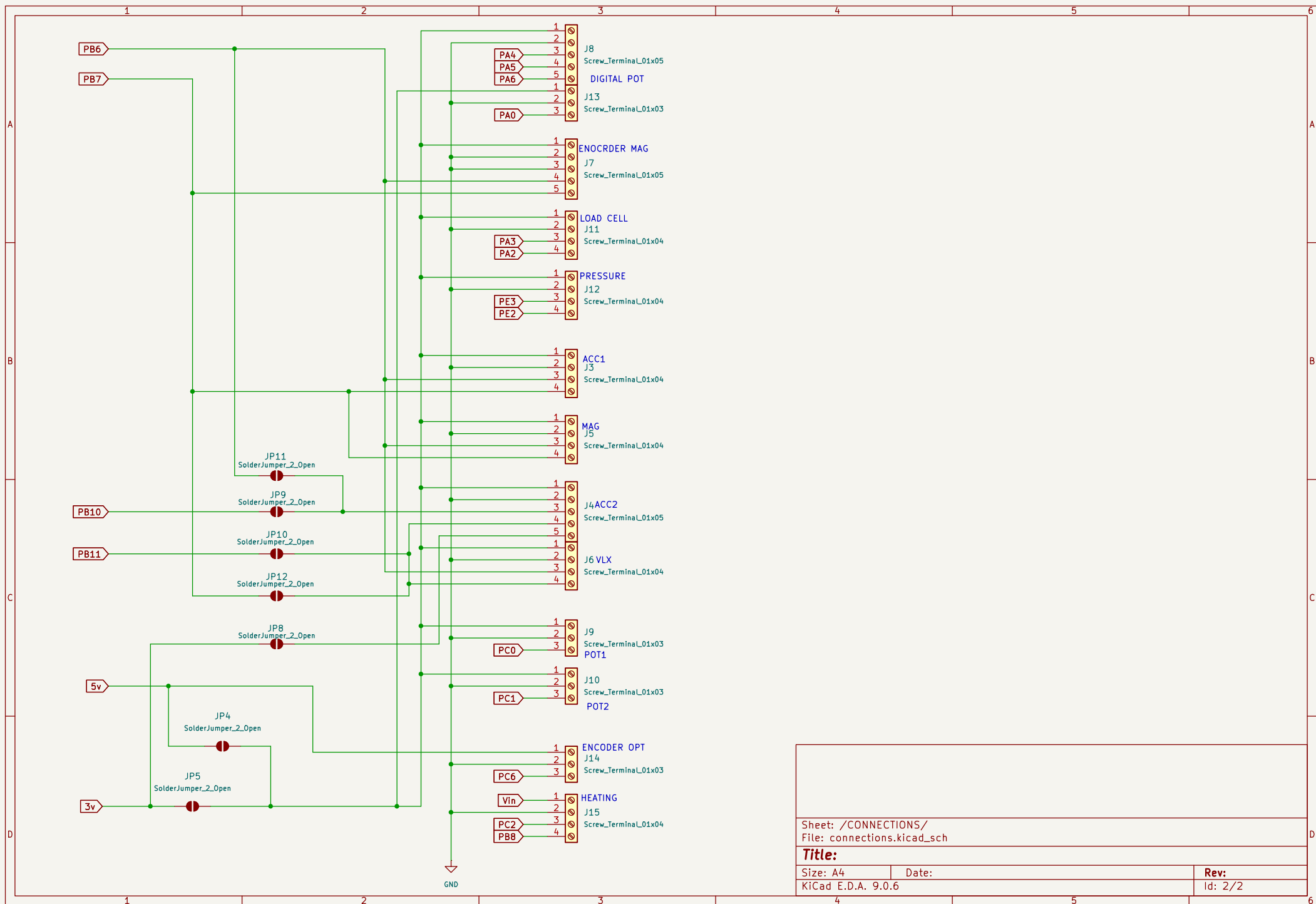
Diagnostic visual feedback array. Eight discrete LEDs with dedicated current-limiting resistors, mapped directly to STM32 GPIO pins for real-time firmware state visualization and debugging.

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